

Dynamic system simulation of small satellite projects [☆]

Matthias Raif^{a,*}, Ulrich Walter^a, Jasper Bouwmeester^b

^a TU Muenchen, Institute of Astronautics, Garching, Germany

^b Delft University of Technology, Chair of Space Systems Engineering, Delft, Netherlands

ARTICLE INFO

Article history:

Received 31 January 2010

Received in revised form

7 June 2010

Accepted 20 June 2010

Keywords:

System simulation

Model-based development

Nanosatellites

ABSTRACT

A prerequisite to accomplish a system simulation is to have a system model holding all necessary project information in a centralized repository that can be accessed and edited by all parties involved. At the Institute of Astronautics of the Technische Universitaet Muenchen a modular approach for modeling and dynamic simulation of satellite systems has been developed called dynamic system simulation (DySyS). DySyS is based on the platform independent description language SysML to model a small satellite project with respect to the system composition and dynamic behavior. A library of specific building blocks and possible relations between these blocks have been developed. From this library a system model of the satellite of interest can be created. A mapping into a C++ simulation allows the creation of an executable system model with which simulations are performed to observe the dynamic behavior of the satellite. In this paper DySyS is used to model and simulate the dynamic behavior of small satellites, because small satellite projects can act as a precursor to demonstrate the feasibility of a system model since they are less complex compared to a large scale satellite project.

© 2010 Elsevier Ltd. All rights reserved.

1. Introduction

A dynamic system simulation can greatly improve the development process of a satellite. Computational simulations can be performed in an early stage of the development process to gain deeper insights into the system performance. Without having any hardware under test, design flaws can be detected and eradicated. With a dynamic simulation it is possible to test the spacecraft design with respect to the requirements to ensure that the performance will satisfy all needs. As a predictive tool a system simulation can be used to evaluate different design alternatives in a trade study. It also can be used to size components of the satellite's subsystems. By varying the

parameter in question several simulations can be made. The results of the different simulations can be compared and conclusions are drawn to derive the best design.

System simulation is a very special form of simulation. It incorporates all the different engineering disciplines into one system model. This model is used in a simulation to analyze the behavior of the complete system. During satellite development a wide range of simulation tools is available to conduct simulations for any special purpose, as there are for example: mission analysis and orbit simulation, structural and thermal analysis, or attitude and control simulation. All these tools are specialized on a certain aspect of the system. None of these tools is able to model and simulate the entire system even though many system details are similar in these separated simulations. In contrast a system simulation will incorporate many aspects into one simulation. Thus it comprises the interactions and influences between the different system aspects and its corresponding dynamic behavior. The drawback of a system simulation is the complexity of the

[☆] This paper was presented during the 60th IAC in Daejeon.

* Corresponding author.

E-mail addresses: m.raif@tum.de (M. Raif), u.walter@tum.de (U. Walter), jasper.bouwmeester@tudelft.nl (J. Bouwmeester).

model. To reduce computational efforts one has to reduce the fidelity and depth of model detail in some aspects compared to dedicated simulation tools. So system simulation cannot completely substitute all other simulations but it can certainly contribute as a valuable supplement.

In this paper we concentrate on the use of dynamic system simulation for small satellite projects, especially in the nanosatellite size. The method described is used to predict the system performance in the early design phase of the project. Therefore it uses simple models to limit complexity. The method is modular and adaptable enough that it can also be used in later phases. The modularity also permits to reuse the developed models in other projects. In fact special attention was paid to the modularity and reusability of the model. In the future there shall be a comprehensive library of building blocks from which a satellite project can be virtually assembled and simulated.

2. System dynamic

The purpose of a dynamic system simulation is to predict or validate the dynamic behavior of a system. This is done by running a system model representing of the real system on a computer. The dynamic behavior is defined as the evolution of all system states over time. The system model has to reproduce this temporal change in an appropriate way. Certainly not all real effects can be included in the system model. Simplifications and abstraction have to be made. To evaluate the system performance of a spacecraft design the most important dynamic behaviors are the kinetic and attitude behavior, the electrical behavior (generation, distribution, and consumption of electrical energy), the thermal behavior (temperature distribution) and the communication behavior (inside and outside the spacecraft). The dynamic behavior of a spacecraft is best described by differential equations specifying the change of system states over time. For the dynamic simulation these differential equations have to be solved and evaluated. This is done by numerical methods.

As only the overall dynamic behavior of the spacecraft system is of interest some simplifications on the model are made:

- Only simple differential equations are evaluated. No partial differential equations are used.
- Forces on structural elements cause no deformation.
- No strain or deformation due to thermal effects.
- Hard radiation and their effects on electrical devices are not modeled (except for solar cells).
- Electromagnetic compatibility of electrical devices is not considered.
- Only Earth orbiting satellites can be simulated at the moment (due to the chosen coordinate frame).

2.1. Kinetic and attitude behavior

The movement and pointing of the spacecraft is evaluated with the help of two coordinate systems.

The system-states position and velocity are given in the Earth centered inertial coordinate system (ECI) (see Fig. 1). The origin of the ECI coordinate system is located at the Earth's center. The x-axis lies on the intersection of equatorial plane and the ecliptic, pointing to the first point of aries. The z-axis lies on Earth's rotational axis pointing northwards. The y-axis completes the coordinate system to a right handed one.

The second coordinate system is a body fixed coordinate system with its origin in the center of mass of the spacecraft.

The rate of change of the spacecraft's position r and velocity v can be expressed by two ordinary differential equations of first order:

$$\frac{dv}{dt} = a_{\text{extern}} + \frac{F_{\text{total}}}{m_{\text{system}}} \quad (1)$$

$$\frac{dr}{dt} = v \quad (2)$$

Here a_{extern} are the external accelerations acting on the system, F_{total} is the sum of all acting forces and m_{system} is the mass of the system.

The system-state attitude is given as a quaternion describing the orientation of the body fixed coordinate system with respect to the inertial coordinate system. The system state angular velocity is given as the angular rates about these body axes.

The rotational movement is described by the rate of change of the angular velocity ω

$$\frac{d\omega}{dt} = I^{-1}(T_{\text{total}} - \omega \times h_{\text{total}}) \quad (3)$$

with I as the moment of inertia of the rigid body, the sum of all acting torques T_{total} and the total angular momentum h_{total} of the system. The rate of change of attitude is given by

$$\frac{dq}{dt} = \frac{1}{2} \Omega q \quad (4)$$

$$\Omega = \begin{bmatrix} 0 & -\omega_1 & -\omega_2 & -\omega_3 \\ \omega_1 & 0 & \omega_3 & -\omega_2 \\ \omega_2 & -\omega_3 & 0 & \omega_1 \\ \omega_3 & \omega_2 & -\omega_1 & 0 \end{bmatrix} \quad (5)$$

with the quaternion q and the angular velocity matrix Ω .

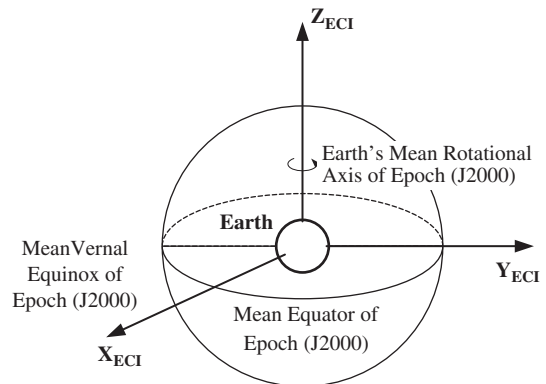


Fig. 1. Definition of the Earth centered inertial coordinate system (ECI).

2.2. Electrical behavior

The electrical behavior describes the generation, storage and consumption of electrical energy inside the spacecraft.

Most of all spacecraft rely on solar cells for the generation of electrical energy. The production rate of solar cells is mainly dependent on the incidence angle of the incoming sunlight, i.e. the attitude of the spacecraft.

The storage of electrical energy is solved by chemo-electrical batteries. The rate of change of current capacity of a battery can be described by a simple differential equation:

$$\frac{dQ}{dt} = P_{\text{total}} \quad (6)$$

with Q as the stored electrical energy and P_{total} as the sum of all generated and consumed electrical power, including all conversion losses on the path.

2.3. Thermal behavior

The thermal behavior describes the temperature distribution inside the spacecraft. As we do not consider partial differential equations only a lumped parameter

model comes into consideration. Here several isothermal nodes exchange thermal energy by conduction and radiation. The number of nodes should be limited to minimize computational effort.

The conductive heat exchange on one node i is described by

$$\dot{Q}_c = \sum_j C_{ij}(T_j - T_i) \quad (7)$$

the radiation exchange by

$$\dot{Q}_r = \sigma \sum_j R_{ij}(T_j^4 - T_i^4) \quad (8)$$

where σ is the Stefan–Boltzmann constant.

A hotter neighboring node j ($T_j > T_i$) results in a positive heat flow on node i . C_{ij} describe the conductive heat-transfer coefficients, R_{ij} the radiation heat-transfer coefficients between i and j .

Each node is characterized by its temperature (the state) and the heat capacity. The rate of change of the temperature is given by

$$\frac{dT}{dt} = \frac{1}{mc} \dot{Q}_{\text{total}} \quad (9)$$

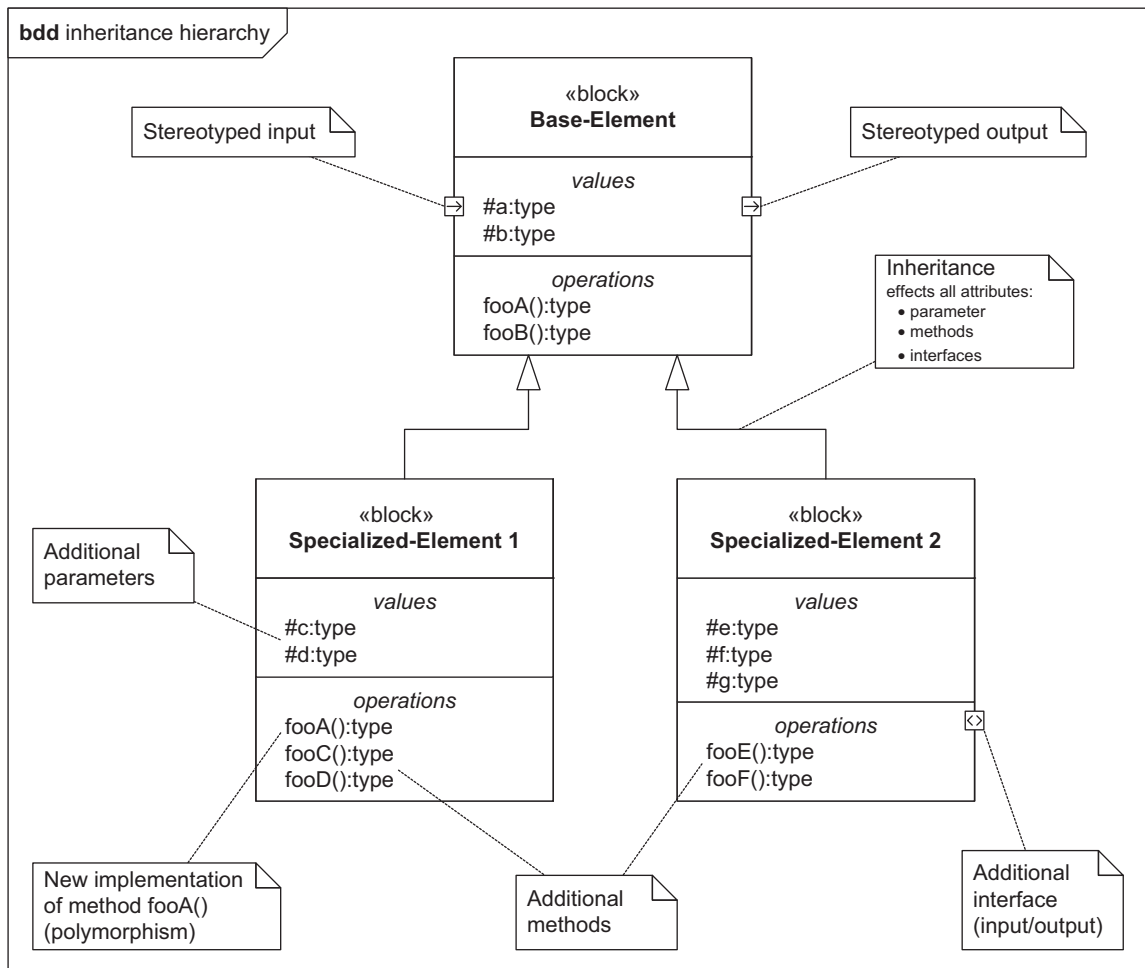


Fig. 2. Concept of inheritance in the element library.

where m is the mass of the node and c the specific heat capacity. $\dot{Q}_{\text{total}}$ is the sum of all heat flows acting on that node

$$\dot{Q}_{\text{total}} = \dot{Q}_c + \dot{Q}_r + \dot{Q}_{\text{diss}} \quad (10)$$

with $\dot{Q}_{\text{diss}}$ as the dissipated heat (e.g. electrical conversion losses) inside the node.

2.4. Communication and data handling behavior

A spacecraft can be seen as an embedded system. A lot of sensors and actuators communicate with an onboard computer via special data handling protocols. For the dynamic system simulation, these data handling have to be modeled only if hardware-in-the-loop tests are involved. In early phases of system design, these analyses can be discarded.

The communication between the spacecraft and the ground station are of interest, especially the quality of the connection and its time of availability. These calculations are highly dependent on the position and attitude of the spacecraft in relation to the ground station.

3. System modeling

The system model for the dynamic simulation should be built very modular to achieve reusability of the model. Each modular element of the model represents a certain part of the system. The element is defined in a generic way so it can represent different types of characteristics. The definition of an element includes the definition of

parameters, interfaces and methods. Interaction between elements is done by relations which can exchange a specified type of item (such as energy, mass, force, data, etc.).

A generic model library of elements is created in a platform independent way using SysML as description language [1]. The modeling method uses the concept of inheritance (Fig. 2) to build up a hierarchy in the model library [2].

All physical elements in the system model are derived from three basic element classes: the system class, the subsystem class and the device class. In addition some properties or objects of the environment are derived from an abstract element class. The complete inheritance hierarchy of the used element classes can be seen in Fig. 3.

A system model is created by decomposing the system into element classes that are described in the library. The required element classes are of all basic element class types (system, subsystem and device). Building a system from bottom up, a subsystem object can be composed of several device objects. A system object is composed of several subsystem objects. These composition relations form the system model hierarchy. An example of a system model hierarchy created from composition-related objects can be seen in Fig. 12.

This modular and adaptive approach has many similarities to the concurrent engineering approach. In both approaches a centralized model of the system is present on which different engineering disciplines can work concurrently. The concurrent engineering approach is currently used to support the very early conceptual design phases of a spacecraft system [3]. The two

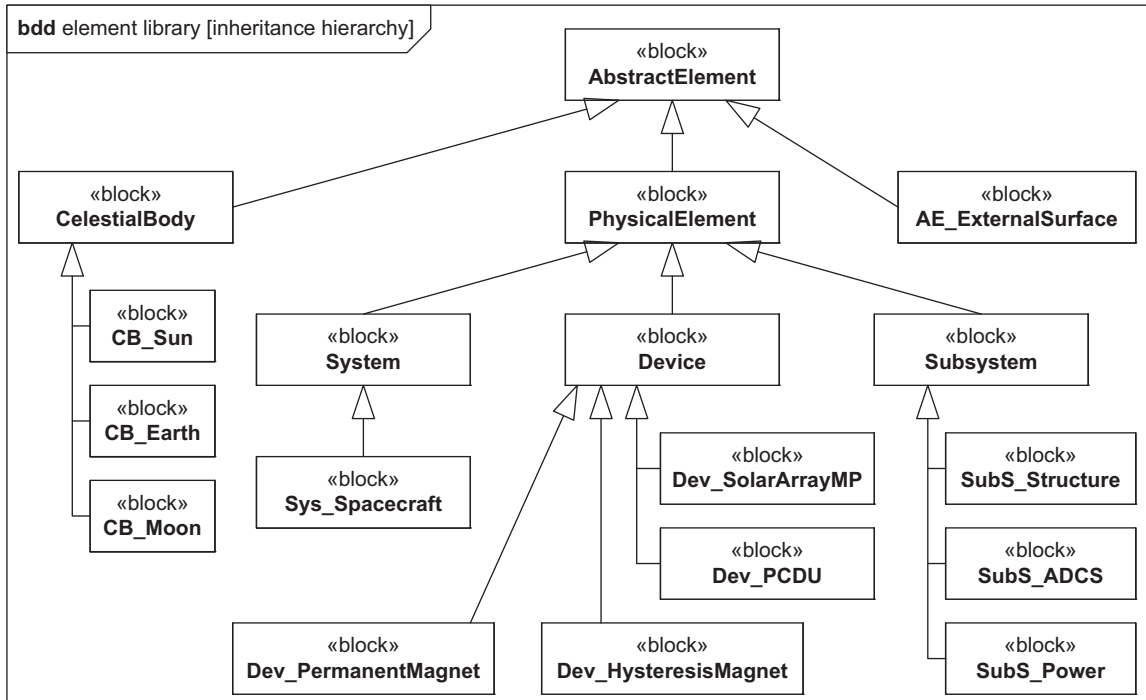


Fig. 3. Complete inheritance diagram of the used library element classes.

approaches can even be combined to evolve a system model from the static representation of a concurrent engineering facility to a system model which is capable to be simulated dynamically. One has to consider that more information is needed for the dynamic simulation. This additional information has to be added during the evolution process from a static to a dynamic model representation.

4. Model library

The following paragraphs give a detailed explanation on the element classes used in this paper. The SysML descriptions shown here are shortened to the structural definition only. The functionality and behavior is explained in the text and with formulas. In the developed modeling method this information is also described by SysML diagrams (parametric diagram, state chart diagram).

4.1. Environment

The environment of a spacecraft is mainly defined by the celestial bodies in its vicinity. Three different celestial bodies have been considered: The Sun, the Earth and the

Moon. All three bodies are based on a common more general element class, in which all parameters, interfaces and methods are declared. The specific characteristic is done in the three specialized element classes (see Fig. 4).

The state of the celestial bodies, i.e. their position and attitude in the ECI coordinate system is determined by approximations given in [4].

The influence of a celestial body on the kinetics of a spacecraft is mainly given by gravitational effects. The gravitational attraction on a spacecraft can be divided into separate fractions. The main attraction of a central celestial body is dependent on the distance and can be calculated as acceleration by

$$\mathbf{a}_{\text{grav}} = -\mathbf{r} \frac{\mu}{|\mathbf{r}|^3} \quad (11)$$

where $\mathbf{r}$ is the radius vector from the celestial body center and μ its gravitational parameter. In addition to this main attraction, some perturbations have to be considered. For a spacecraft in Earth orbit the variations of Earth's gravitational field can be calculated by a spherical harmonic expansion term B as part of the total gravitational potential U . The acceleration is obtained by determining the gradient of the gravitational potential ∇U . In spherical coordinates the perturbation part of

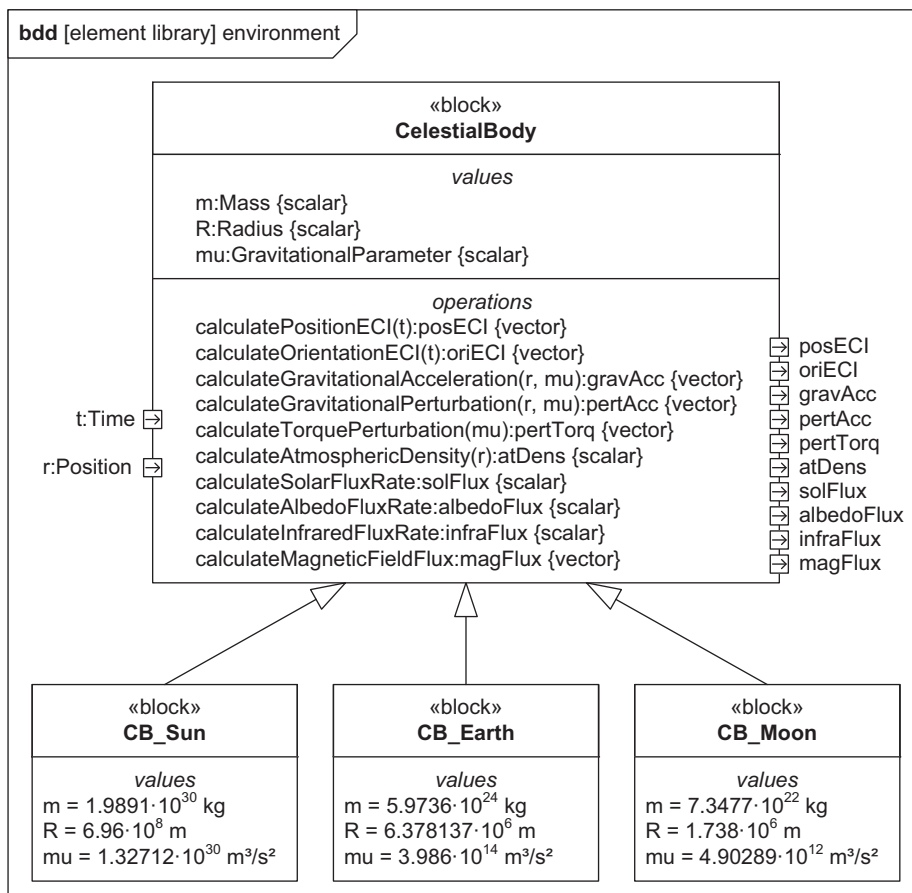


Fig. 4. Definition of the element class celestial body.

Earth's gravitational potential can be expressed by

$$B(r, \theta, \phi) = \frac{\mu_{\oplus}}{r} \left\{ \sum_{n=2}^{\infty} \left[\left(\frac{R_{\oplus}}{r} \right)^n J_n P_{n0}(\cos \theta) + \sum_{m=1}^n \left(\frac{R_{\oplus}}{r} \right)^n (C_{nm} \cos m\phi + S_{nm} \sin m\phi) P_{nm}(\cos \theta) \right] \right\} \quad (12)$$

Here, $R_{\oplus}$ is the radius of the Earth, J_n are the zonal harmonic coefficients, P_{nm} are Legendre polynomials, and C_{nm} and S_{nm} are tesseral harmonic coefficients for $n \neq m$ and sectoral harmonics for $n = m$ [5]. The gravitational perturbation by the Sun and the Moon can be calculated as third-body influence

$$\mathbf{a}_{\text{pert}} = -\mathbf{d} \frac{\mu}{|\mathbf{d}|^3} \quad (13)$$

for each celestial body considered, where $\mathbf{d}$ is the distance vector between the spacecraft and the celestial body [6]. The total acceleration on a spacecraft is simply the sum of all these influences.

The attitude of a spacecraft is also influenced by gravitation (gravity gradient) but further more by atmospheric drag, Earth's magnetic field interaction and solar radiation pressure.

The effect of gravity gradient perturbation is covered by the causing gravitational parameter μ . The calculation of the resulting torque is done inside the spacecraft model. The other influences of the environment on the spacecraft are given by the atmospheric density [7], the magnetic field flux [8], and the radiation (solar and infrared) at the spacecraft's position [9].

Solar and albedo radiation influences the power generation of solar cells. These two together with the infrared radiation also affect the thermal behavior of the satellite.

Table 1 shows the mapping of the celestial body outputs to the corresponding model used to calculate the physical effect.

4.2. Spacecraft system

The spacecraft system element comprises all parameters, states, interfaces and methods that are unique to the system as a whole. States of the spacecraft system, like position, velocity and so on, also apply to all elements, the

system is composed of. The system hierarchy is used to determine all physical properties on system level. Thus e.g. the total mass of the spacecraft is calculated by the sum of all its component masses.

The rate of change of position, velocity, attitude and angular velocity is determined according to Eqs. (1)–(4). The total external acceleration $\mathbf{a}_{\text{extern}}$ is given by

$$\mathbf{a}_{\text{extern}} = \mathbf{a}_{\text{grav}} + \mathbf{a}_{\text{pert}} \quad (14)$$

The total torque on the spacecraft system $\mathbf{T}_{\text{total}}$ is given by

$$\mathbf{T}_{\text{total}} = \sum \mathbf{T}_i + \sum (\mathbf{F}_j \times \mathbf{O}_j) + \mathbf{T}_{\text{gg}} \quad (15)$$

Here $\mathbf{T}_i$ are all input torques, $\mathbf{F}_j$ are all input forces and $\mathbf{O}_j$ the corresponding origin vector, pointing from the center of mass to the origin of the force vector. $\mathbf{T}_{\text{gg}}$ is the gravity gradient torque determined by

$$\mathbf{T}_{\text{gg}} = \frac{3\mu}{r^5} [\mathbf{r} \times (\mathbf{I} \cdot \mathbf{r})] \quad (16)$$

Interfaces exist in order to connect the spacecraft system with the environment. The complete definition of the spacecraft system can be seen in Fig. 5.

4.3. Subsystems

The spacecraft's subsystems usually have no function themselves; they serve as an abstract container to accumulate devices and equipment which together fulfill the function. Also in the system model, subsystems serve as container, where devices can be assigned to, just to keep a clear view.

The only exception in the system model is made for the structure subsystem. This subsystem serves as the attachment point for all devices in the system. Forces and torques produced by devices are transferred to the structure subsystem. This resembles the fact that hardware is attached to the structure in reality. The total force and torque is transferred to the system level where it is evaluated further.

4.4. Devices

The functionality of the spacecraft system is enabled by the devices the spacecraft is composed of. This section will describe the models of the devices used in the example scenario. Each device is assigned to a subsystem according to the functionality it provides.

4.4.1. Attitude

The attitude of a spacecraft system is altered by the devices grouped into the attitude control subsystem (ACS). For nanosatellites this is often done by passive magnetic attitude stabilization (PMAS) [10] (definition of elements see Fig. 7).

For this a permanent magnet is placed inside the satellite which interferes with Earth's magnetic field, producing a vague control torque. Oscillations can be damped by hysteresis magnets placed perpendicular to the permanent magnet. The attitude about the local magnetic field vector of Earth's field cannot be influenced.

Table 1
Mathematical models used in the different celestial body elements.

Output	Used mathematical model in element		
	Sun	Earth	Moon
posECI	$f(\text{time})$	0	$f(\text{time})$
oriECI	–	$f(\text{time})$	–
gravAcc	–	Eq. (11)	–
pertAcc	Eq. (13)	Eq. (12)	Eq. (13)
pertTorq	–	$\mu_{\oplus}$	–
atDens	–	Gill	–
solFlux	SMAD	–	–
albedoFlux	–	SMAD	–
infraFlux	–	SMAD	–
magFlux	–	Simple dipole/IGRF 10th	–

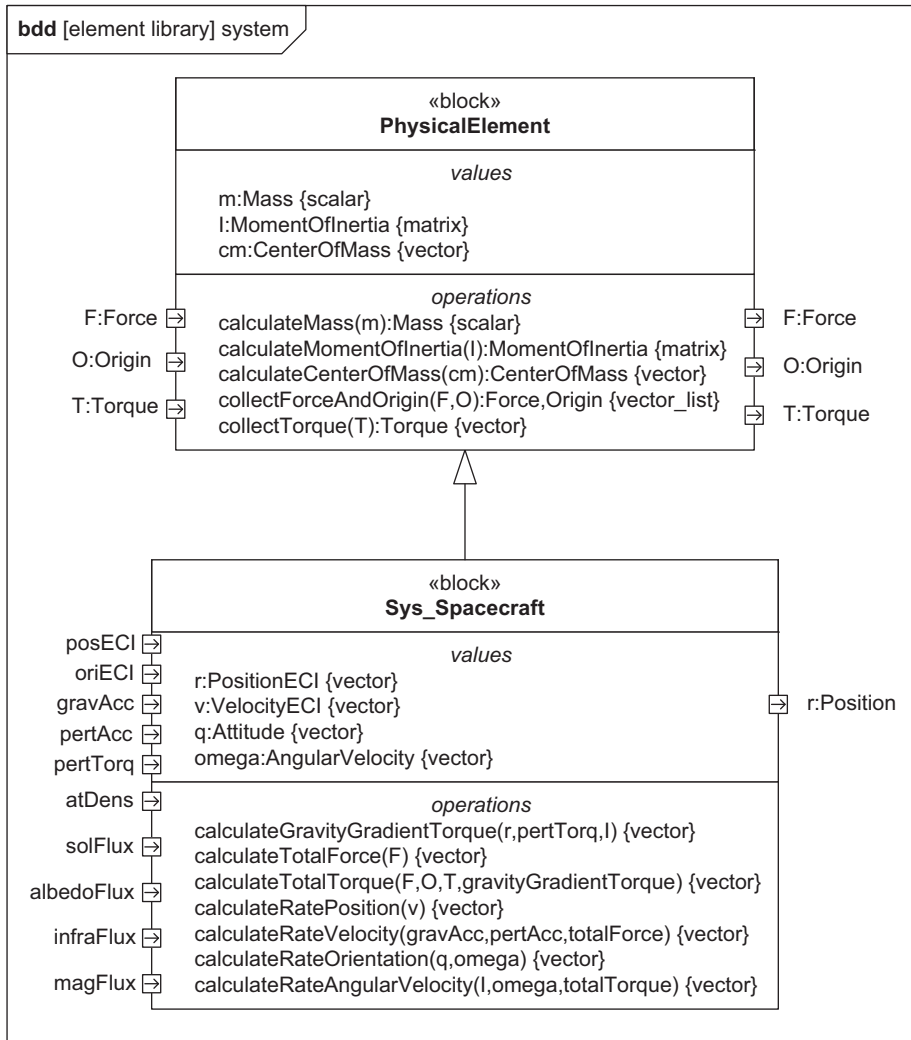


Fig. 5. Definition of the spacecraft system element class.

The torque produced by a magnet is calculated by

$$\mathbf{T}_{\text{mag}} = \mathbf{m} \times \mathbf{B}_{\text{ext}} \quad (17)$$

here $\mathbf{B}_{\text{ext}}$ is Earth's magnetic field flux and $\mathbf{m}$ the magnetic dipole of the magnet given by

$$\mathbf{m} = \mathbf{M}V \quad (18)$$

with the volume of the magnetic material V . The magnetization $\mathbf{M}$ of the magnetic material is the result of the external magnetic field and the material properties.

$$\mathbf{M} = \frac{\mathbf{B}_{\text{mat}}}{\mu_0} - \mathbf{H}_{\text{ext}} \quad (19)$$

The magnetic flux density $\mathbf{B}_{\text{mat}}$ of the magnetic material is altered by the external magnetic field $\mathbf{H}_{\text{ext}} = \mathbf{B}_{\text{ext}}/\mu_0$. The value for the magnetic flux density is given by the B – H curve of the magnetic material. The magnetic field strength in low Earth orbit is about up to 30 A/m, which covers the complete saturation cycle of a hysteresis material shown in Fig. 6, if the material is completely rotated in the external magnetic field. The

magnetic flux density of a permanent magnet is hardly altered by this external field, due to the high coercivity value of the material.

The hysteresis characteristic is modeled with an approximation developed by Tellinen [11] which is also able to simulate minor loops. The model determines the change of magnetic flux density b in the direction of the longitudinal axis of the hysteresis rod due to the applied effective external magnetic field strength h (the effective field strength h is the projection of the external field onto the longitudinal axis of the hysteresis material). For increasing external field strength ($dh > 0$) the change in magnetic flux density is given by

$$\frac{db}{dh} = \mu_0 + \frac{b_{h-}(h) - b}{b_{h-}(h) - b_{h+}(h)} \left[\frac{db_{h+}(h)}{dh} - \mu_0 \right] \quad (20)$$

and for decreasing external field strength ($dh < 0$)

$$\frac{db}{dh} = \mu_0 + \frac{b - b_{h-}(h)}{b_{h-}(h) - b_{h+}(h)} \left[\frac{db_{h-}(h)}{dh} - \mu_0 \right] \quad (21)$$

Here μ_0 is the permeability of vacuum. The slope of $b(h)$ is approximated by an inverse tangent function resulting in the rising slope

$$b_{h+}(h) = 2 \frac{B_s}{\pi} \arctan\left((h-H_c) \frac{1}{H_c} \tan\left(\frac{B_r \pi}{B_s}\right)\right) \quad (22)$$

and the falling slope

$$b_{h-}(h) = 2 \frac{B_s}{\pi} \arctan\left((h+H_c) \frac{1}{H_c} \tan\left(\frac{B_r \pi}{B_s}\right)\right) \quad (23)$$

The derivatives are:

$$\frac{db_{h+}}{dh}(h) = 2 \frac{B_s}{\pi} \frac{1}{1 + ((h-H_c)(1/H_c) \tan((B_r/B_s)(\pi/2)))^2} \frac{1}{H_c} \tan\left(\frac{B_r \pi}{B_s}\right) \quad (24)$$

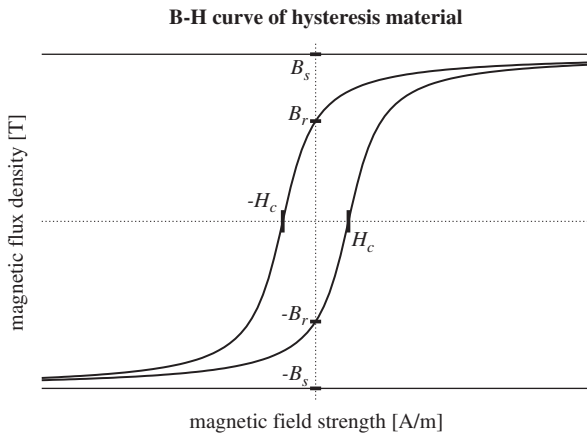


Fig. 6. Saturation loop of a hysteresis material.

$$\frac{db_{h-}}{dh}(h) = 2 \frac{B_s}{\pi} \frac{1}{1 + ((h+H_c)(1/H_c) \tan((B_r/B_s)(\pi/2)))^2} \frac{1}{H_c} \tan\left(\frac{B_r \pi}{B_s}\right) \quad (25)$$

Here B_s is the saturation flux density, B_r the remanence and H_c the coercive force of the magnetic material.

4.4.2. Power

The power subsystem of the investigated nanosatellites is composed of solar cells arranged to small arrays, a power conditioning unit and mostly a battery. For a first analysis of the power behavior of the satellite it is sufficient to look into the power generation of the solar arrays (Fig. 8). The power output of a solar array is

$$P_{el} = \frac{S}{S_{\odot}} n U(T) I(T) \quad (26)$$

Here S is the incoming solar flux, composed of solar radiation (dependent on the sun-satellite distance) and albedo radiation, $S_{\odot}$ is the solar constant (1368 W/m^2), n is the number of solar cells in the array and U are the voltage and I the current of the solar cell. It is assumed that the solar cells work in the maximum power point. Voltage and current are dependent on the temperature of the solar cell resulting in

$$U(T) = U_{MP} + \frac{dU_{MP}}{dT} (T - T_0) \quad (27)$$

$$I(T) = I_{MP} + \frac{dI_{MP}}{dT} (T - T_0) \quad (28)$$

with U_{MP} as the voltage and I_{MP} as the current at maximum power point, dU_{MP}/dT and dI_{MP}/dT are the temperature coefficients, T is the actual array temperature and T_0 is the temperature basis for the

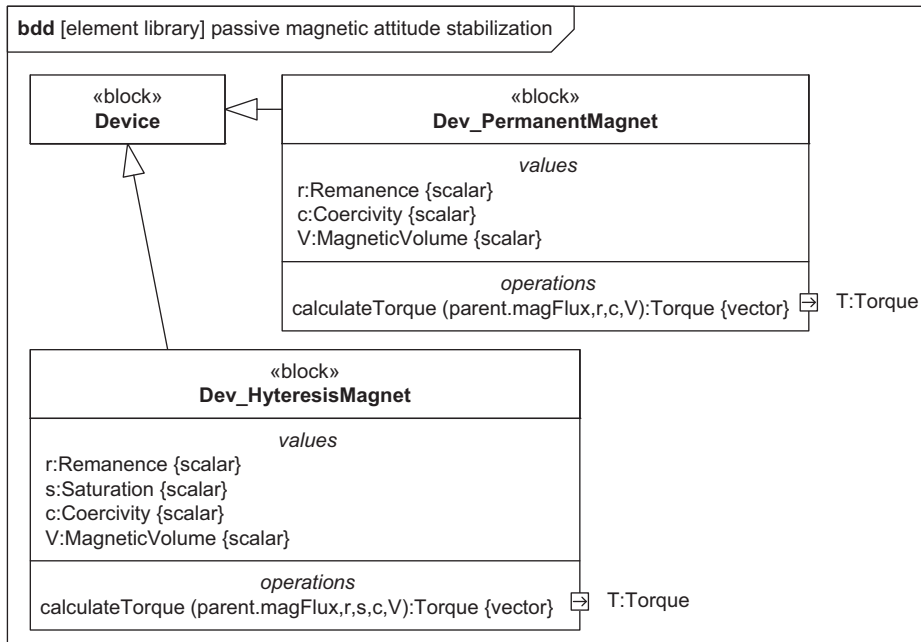


Fig. 7. Definition of the magnet device element classes.

coefficients. Lifetime degradation of the solar cells has been ignored.

4.5. Abstract elements

Besides the physical elements described so far, some items not related to a physical representation of the real world exist. These abstract elements are used to give additional functionality in the system model or to hold parameters which are not related to a certain physical object.

The abstract element presented here is the external surface element class (see Fig. 9). This provides additional properties and interfaces to already existing physical elements. The external surface element class serves as an interface between system elements and the environment. It is responsible for the calculation of solar pressure force and atmospheric drag. It calculates the solar flux on the surface dependent on the orientation of the surface and the heat flux coming from and radiated to the environment.

5. Computational simulation

The abstract and platform independent model described in the previous section has to be transformed into an executable form to run a simulation. The transformation is dependent on the target platform. In this case the target platform is a simulation framework implemented

in C++. Other target platforms like MatLab/Simulink are possible.

The simulator is part of the simulation framework and should be independent of the particular system model it simulates. Simulator and system model are integrated to perform the system simulation. The simulator takes care of the execution of all functionalities of the system model and the communication between the elements in the model. It propagates the states of the system elements through time with the help of an integrator. It also serves as the man-machine interface to control and monitor the simulation run.

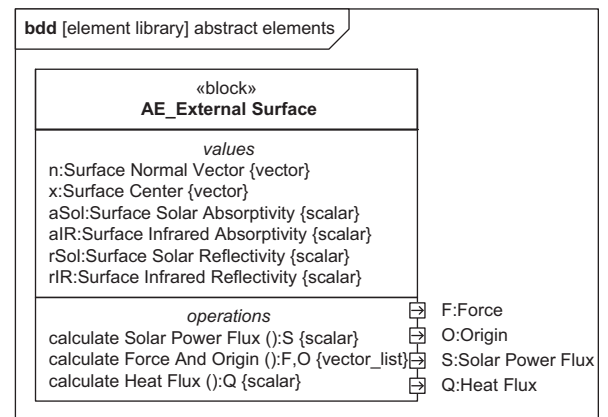


Fig. 9. Definition of the abstract element class external surface.

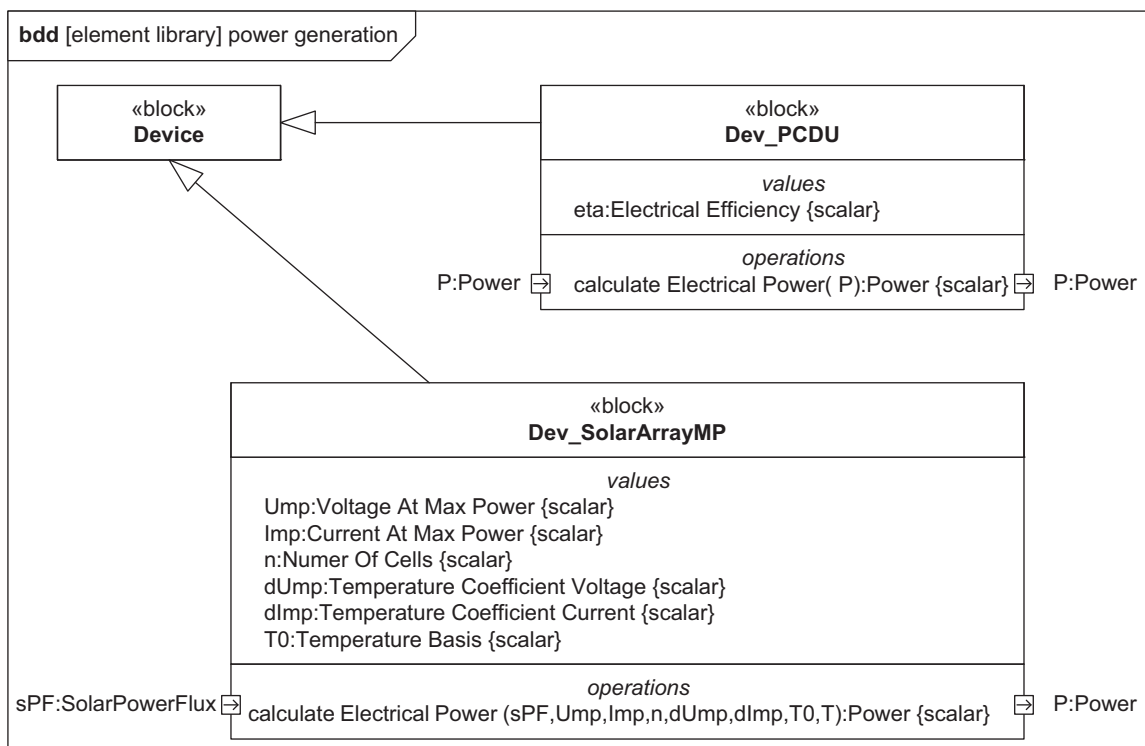


Fig. 8. Definition of the solar array device element class.

In a first step the modeled element classes have to be translated into a C++ implementation. This can be done in a near automated way by code generation techniques out of the SysML representations. The code generator implements the structure (i.e. parameter and method declarations) of the element in a way which the simulator can understand. It also implements services and interfaces which are needed by the simulator to enable communication. The transformation is done by the export functionality of the used modeling tool (v)Sys-ed [12] which was developed at the Institute of Astronautics, TU Munich. The translation process produces a C++ class library of elements which can be used in the simulation framework.

The next step towards a system simulation is to build a system model from the already defined element library. This also can be done graphically with the help of SysML diagrams [2]. These diagrams can be translated into a configuration file which can be read in by the simulator. The simulator builds the simulation model from the C++ element classes in the library according to the configuration file. Afterwards the simulation can start and the simulation model is executed.

The complete workflow from the element description until the simulation model can be seen in Fig. 10.

During simulation the simulator calls the methods of the simulation model in an iterating cycle. If an element has a dynamic state variable, the simulator calls for the derivative of this state variable. After the derivatives of all element state variables are collected the simulator integrates the state variables simultaneously and propagates the new state through the complete simulation model. The integration method can be chosen between an explicit Euler method and a Runge–Kutta 4th order method. The time step for each individual differential equation (i.e. each state variable) can be set as a fraction of the overall simulation frequency. This multi-framing technique avoids problems related to stiffness when the

characteristic time constants for different parts of the model vary significantly.

6. Application

To evaluate the modular modeling process and the simulator results the nanosatellite project Delfi-C³ from Delft University of Technology [13] was chosen as a case study. In [14] most of the information needed to setup a system model was given. The Delfi-C³ satellite is a triple CubeSat configuration with four deployable solar panels and passive magnetic attitude stabilization (Fig. 11). The performance of the power generation and the PMAS is the subject of a system simulation. This first application of the method is only on the attitude and power behavior of

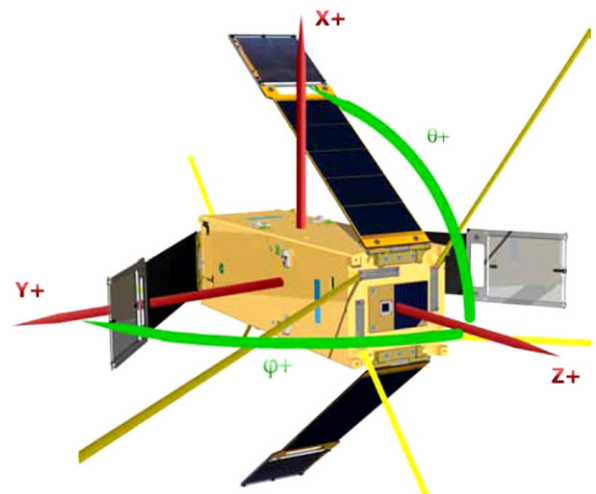


Fig. 11. Configuration of the Delfi-C³ satellite in orbit and the applied coordinate system [14].

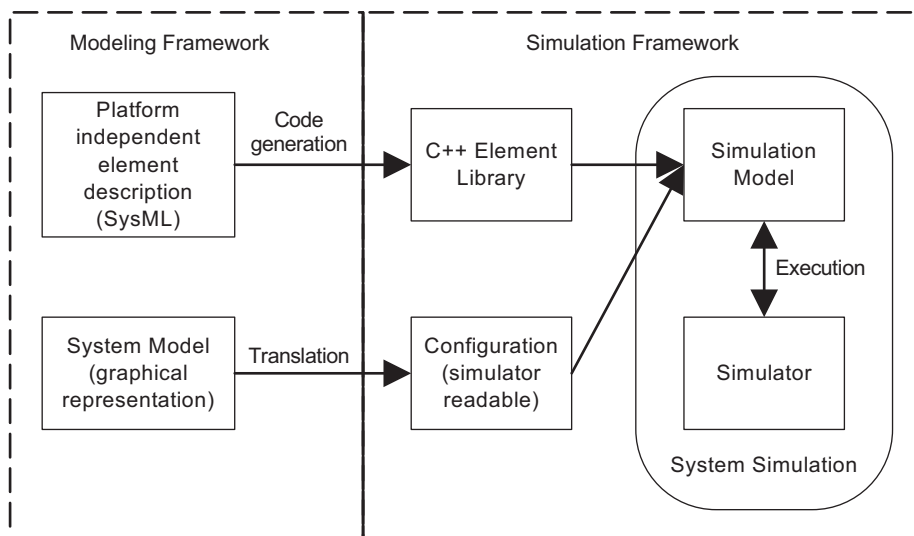


Fig. 10. Workflow for the generation of a system model and its simulation.

Delfi-C³. Thermal and communication behavior is foreseen in the approach and in the model library but not included in the system model of this paper.

The satellite was launched on 28.04.2008 and injected into a sun synchronous orbit with a height of 630 km. The initial rotational rate of $5.1^\circ/\text{s}$ was damped to $0.7^\circ/\text{s}$ after two and a half months of operation (see Fig. 13 for measured values). This corresponds to a depression of $0.05714^\circ/\text{s}$ per day or approximately $0.00386^\circ/\text{s}$ per orbit. The actual power output of the solar array is between 2.4 W at minimum [13] and about 5 W at maximum.

The system model for the simulation is build upon very few basic element classes. The satellite system consists of a structure, an attitude control subsystem (ACS), and a power subsystem. The ACS is composed of a permanent magnet and two hysteresis magnets. The power subsystem comprises the four deployed solar panels with the solar cells, and a power meter, determining the total power generated by all solar panels. The physical properties of the satellite and its components as well as the initial conditions of all system states can be seen in the product tree view system model diagram (see Fig. 12). The satellite's coordinate system origin is located in the geometrical center of its body. The center of mass has a small deviation in z-direction.

The interaction of the system model elements is described in internal block diagrams showing the item flows between the elements. To keep a level of clarity in the modeling process, different views on the system

model can be used, each showing a related functionality. Fig. 14 shows the mechanical view combining all item flows concerning the kinetic and attitude behavior of the satellite. Fig. 15 shows the electric view, combining all item flows concerned with the generation of electrical power.

Solar and atmospheric pressure forces and torques are calculated by taking only the solar panels, the front side and the back side of the satellite into account. The deployed solar panels act as side walls of the satellite and only the front sides of the solar panels contribute to the calculations. A small calculation error appears, as the surfaces of the spacecraft body sides and the backsides of the panels are neglected.

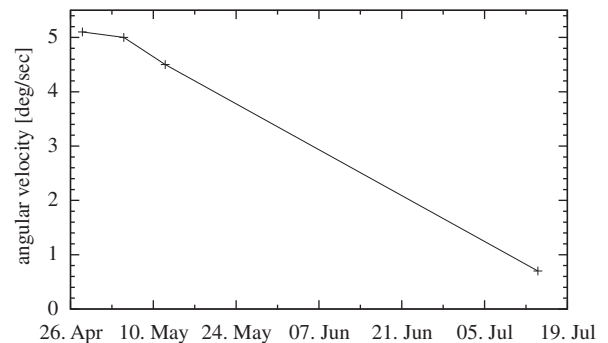


Fig. 13. Measured data on the rotational rate of Delfi-C³.

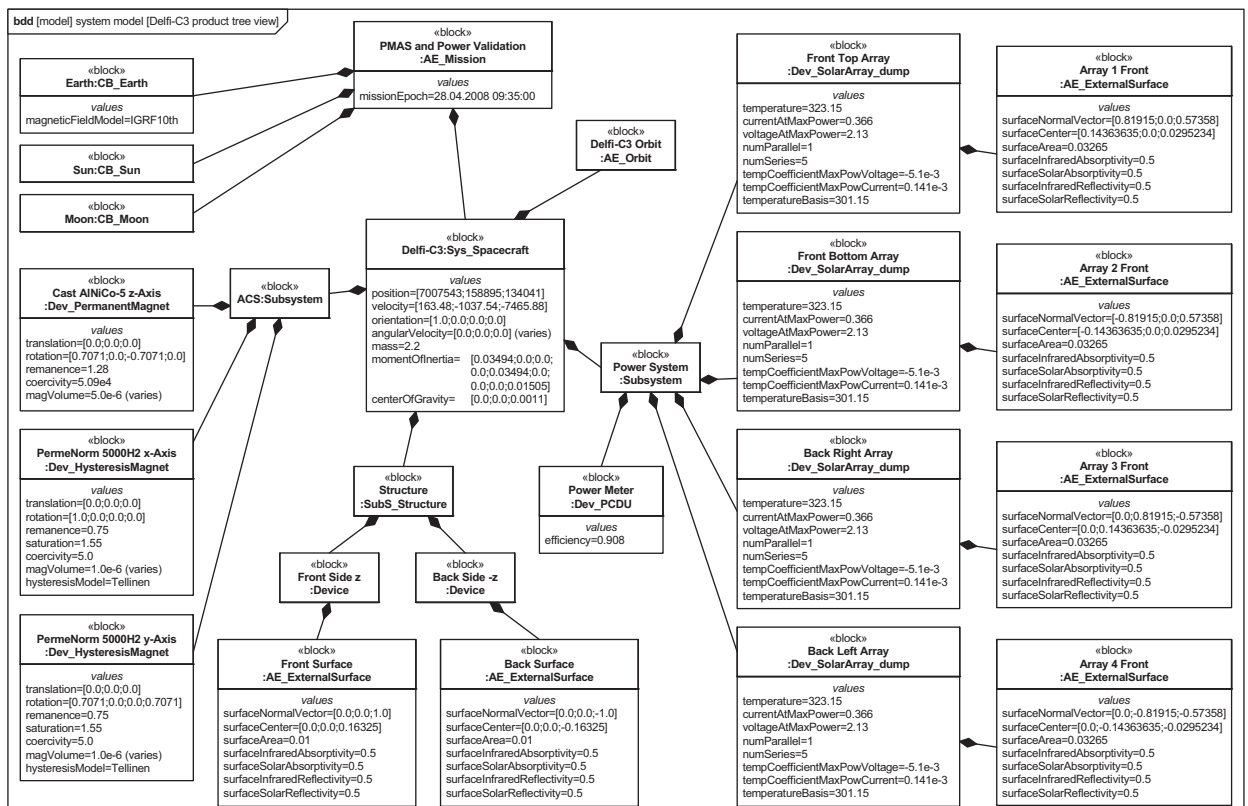


Fig. 12. Product tree view on the system model of the Delfi-C³ satellite to simulate the attitude and power generation behavior.

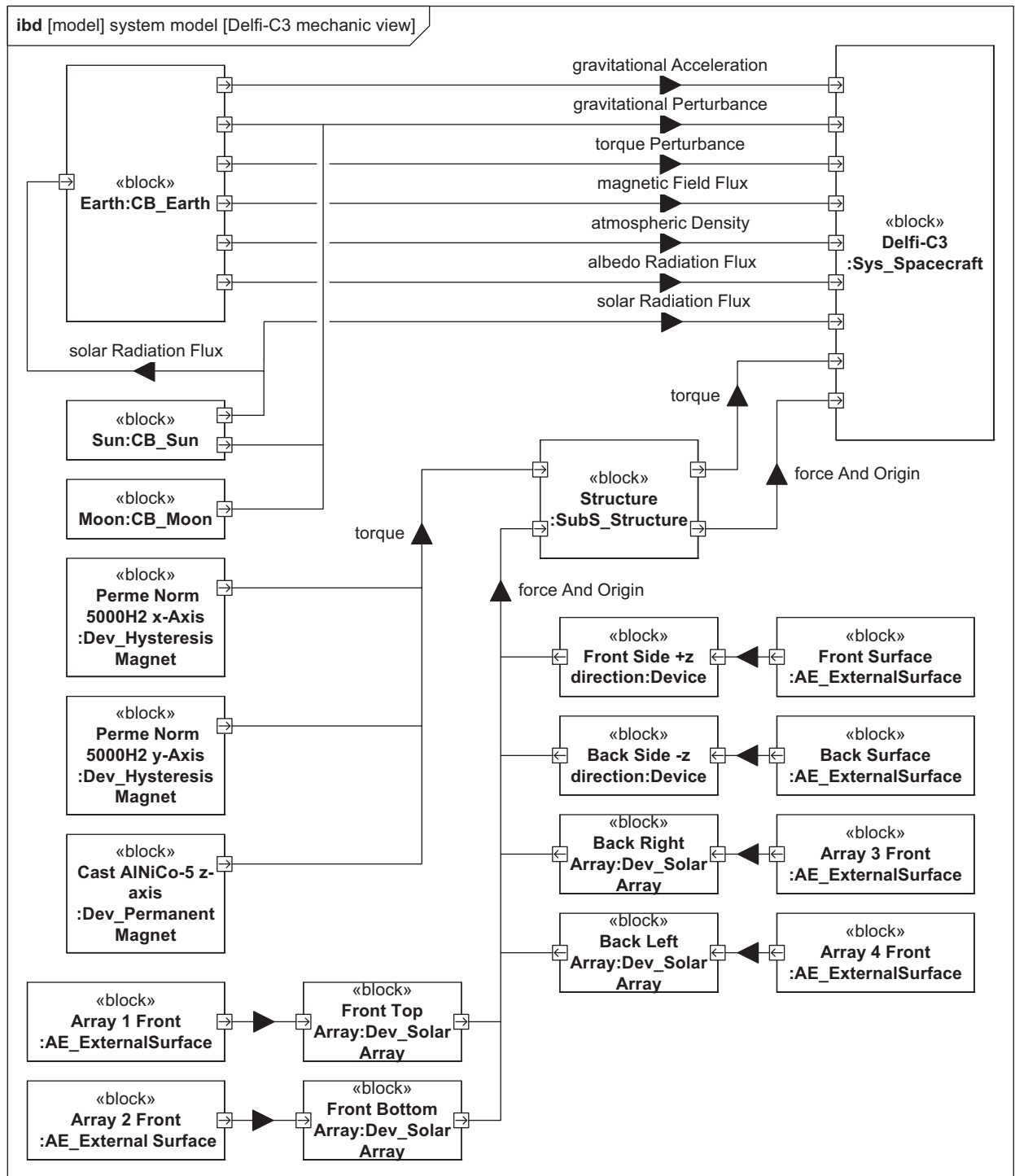


Fig. 14. Mechanic view on the system model.

Solar power flux on the solar arrays is composed of direct solar flux plus albedo flux. Each solar panel calculates the maximum power point power output considering all external effects. Temperature effects are

taken into account but a constant temperature of 323.15 K (50 °C) is used. A power meter device, which simulates the voltage regulator, is responsible for the measuring of all electrical power outputs of the panels.

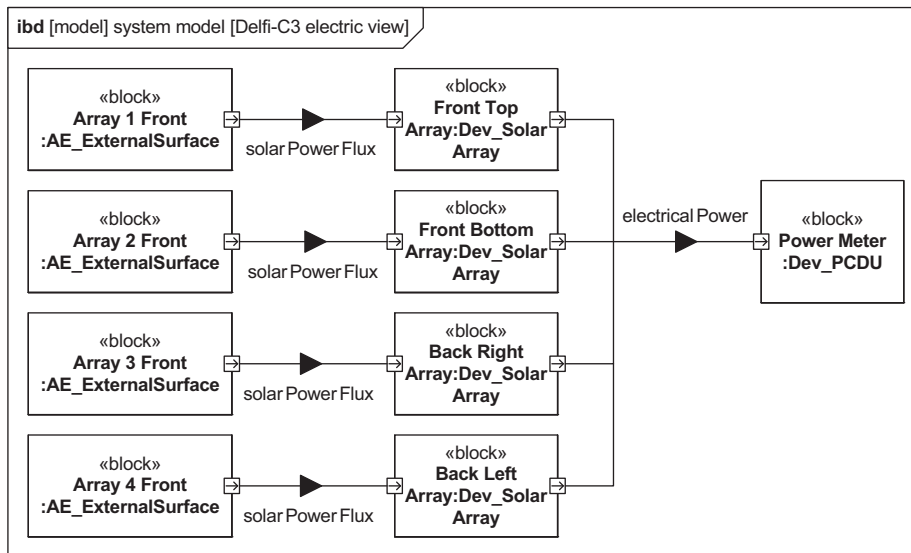


Fig. 15. Electric view on the system model.

6.1. Attitude behavior

The total torque on the system is primarily accounted by the permanent magnet. The magnitude of the total torque is about 5×10^{-5} N m at maximum. The contribution of all other disturbances, aside from the magnetic moment, sums up to some 10^{-7} N m.

Simulation with varying initial conditions show that magnetic lock is achieved within a few orbits. Residual oscillations are present about all axes which are slowly damped by the hysteresis magnets.

Both magnetic materials, the permanent magnet and the hysteresis magnet, contribute to the total system magnetic moment vector and thus to the orientation of the alignment axis [10]. The angle between the permanent magnet axis and the magnetic field lines, which can be seen as the alignment error is dependent on the ratio of the magnetic moment contribution by the permanent and hysteresis magnets. Since the magnetic moment is proportional to the amount of magnetic material, the alignment error is dependent on the mass or volume ratio of the materials. A tradeoff has to be made to find the right ratio. With almost no hysteresis material present, the alignment error can be reduced to almost zero. In this case the damping capability is no longer granted and the residual oscillation will last for a long time.

Two different system model setups for the attitude stabilization system were used in the simulation. The first setup uses an initial guess for the amount of magnetic material with a high ratio of 25:1 of the permanent magnet material to the hysteresis material. The second system model setup reproduces the real amount of magnetic material present in Delfi-C³. The ratio here is 1:5 with more hysteresis material than permanent magnetic material. A comparison between the simulation results and real measurements of the magnetic dampening effects suggests an error in the mathematical model of

the hysteresis material. For this reason an adaption of the model was made and the simulation was redone with the second system model setup.

6.1.1. PMAS setup 1

In the first system model setup, the volume of the permanent magnet is set to 5×10^{-6} m³, the volume of the hysteresis material is 1×10^{-7} m³ on each perpendicular axis. Figs. 16–18 show the prints of the alignment of the permanent magnet axis (z-axis of the satellite body) to the magnetic field lines and to the Nadir vector for the three different initial conditions listed in Table 2 from top to bottom. The initial orientation of the spacecraft was aligned to the ECI coordinate system.

In the first diagram one can see a perfect alignment to the magnetic field lines. The satellite flips twice an orbit at the poles in each case. The second diagram also shows a magnetic lock of the satellite, but due to the initial angular velocity about the z-axis and the mass distribution, the satellite nutates. This results in a deviation of the body axes to the magnetic field axis of about 45°. Also strong residual oscillations are present which are hardly damped. The same effect can be seen in the third diagram. Magnetic lock is established after one orbit, the deviation between body z-axis and magnetic field lines is about 50°.

The oscillations can be seen in Figs. 19–21 showing the progress of the angular velocities. The steady scenario has almost no angular velocity present. In the rotating-z scenarios the angular velocity about the z-axis is exchanged to the other body axes after a short period of time. This is correlated with the magnetic lock. Afterwards the angular velocities stay relatively constant. The bottom diagram shows the other way around. The angular velocity is quickly distributed over the x- and y-axis and some of the angular velocity is transferred to the z-axis. The dampening effect is very

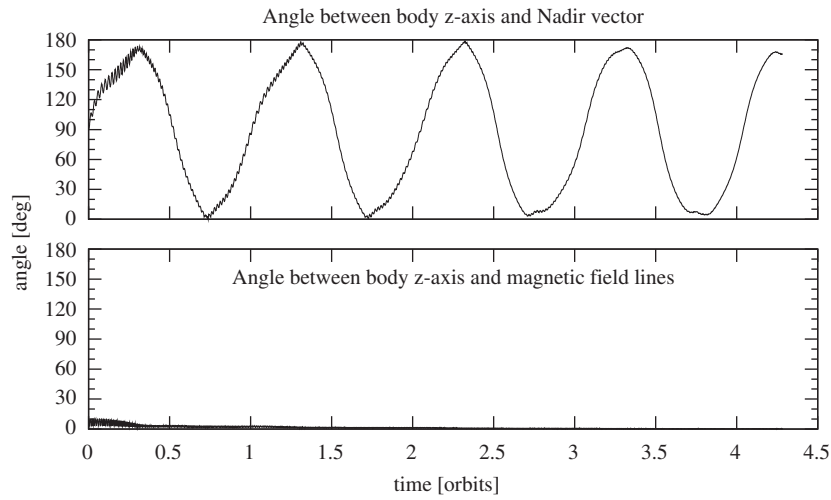


Fig. 16. Alignment of the body z-axis for the steady scenario. $V_{hyst} = 1 \times 10^{-7}$ on x- and y-axis; $V_{mag} = 5 \times 10^{-6}$ on z-axis.

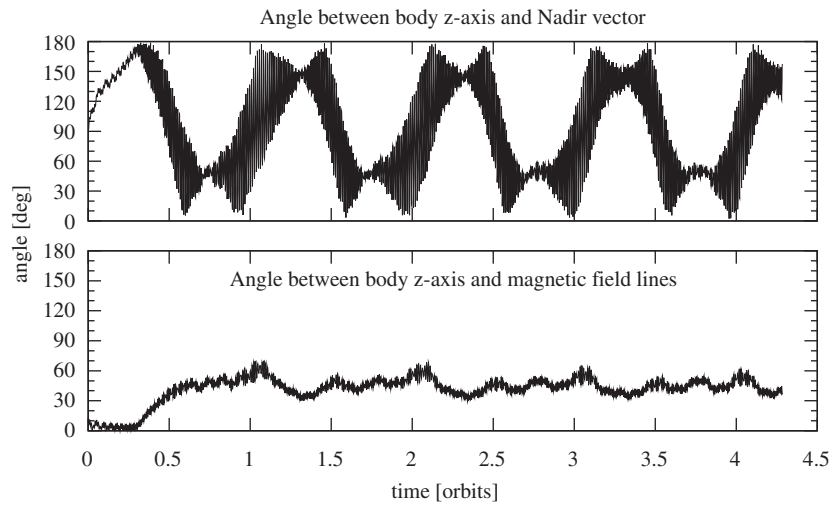


Fig. 17. Rotation z scenario.

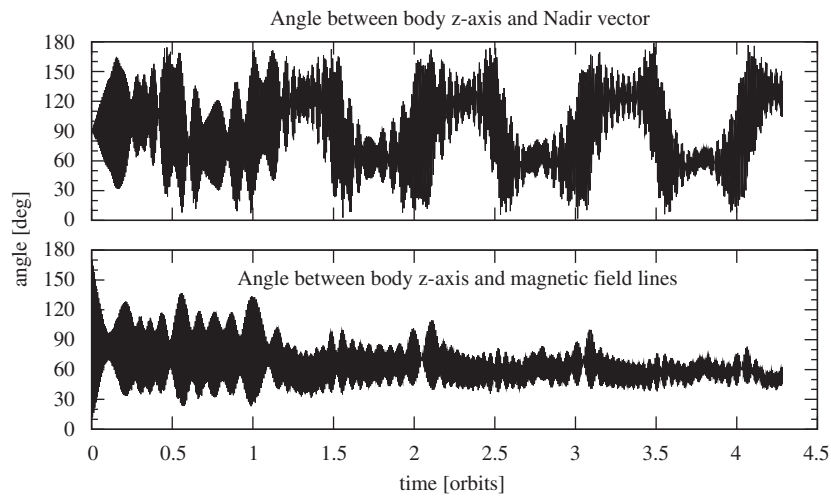


Fig. 18. Rotation x scenario.

weak in all cases. Fig. 22 shows the magnitude of the angular velocity vector for the rotation-x scenario. The decrease in angular velocity is about $0.1653^\circ/\text{s}$ per orbit.

Table 2
Scenarios and initial conditions.

Scenario	Initial angular velocity vector
Steady	$(0 0 0)^T$
Rotation z	$(0 0 0.175)^T$
Rotation x	$(0.175 0 0)^T$

6.1.2. PMAS setup 2

The arbitrary volume of the magnetic materials in the previous setup is now changed to the real values of Delfi-C³. The volume of the permanent magnet is $2.977 \times 10^{-7} \text{ m}^3$, the volume of the hysteresis material is $7.69 \times 10^{-7} \text{ m}^3$ on each perpendicular axis. This is a much different ratio between the two materials, less permanent magnet and more hysteresis material. The satellite is supposed to align not so well to the magnetic field lines. This is desired by the system design because the two experimental payloads of Delfi-C³, a sun sensor and thin film solar cells, should be lighted from different angles. Therefore the attitude control system is supposed

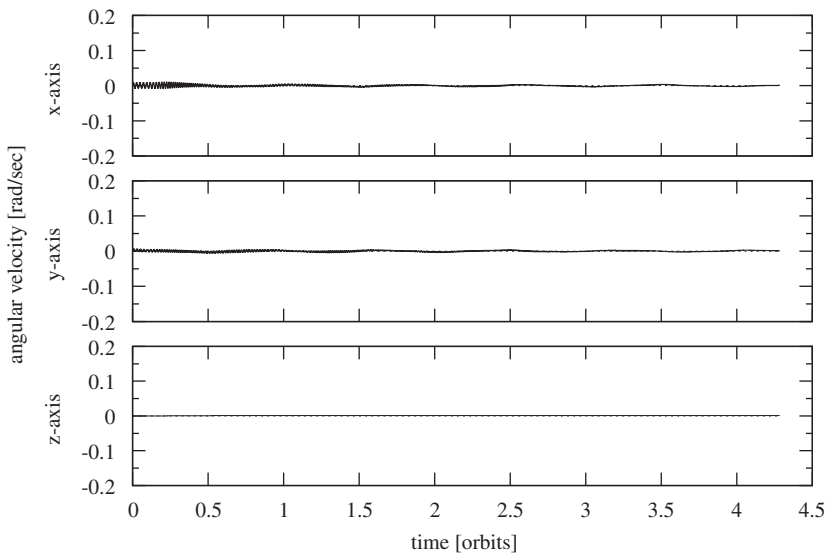


Fig. 19. Angular velocity response for steady scenario.

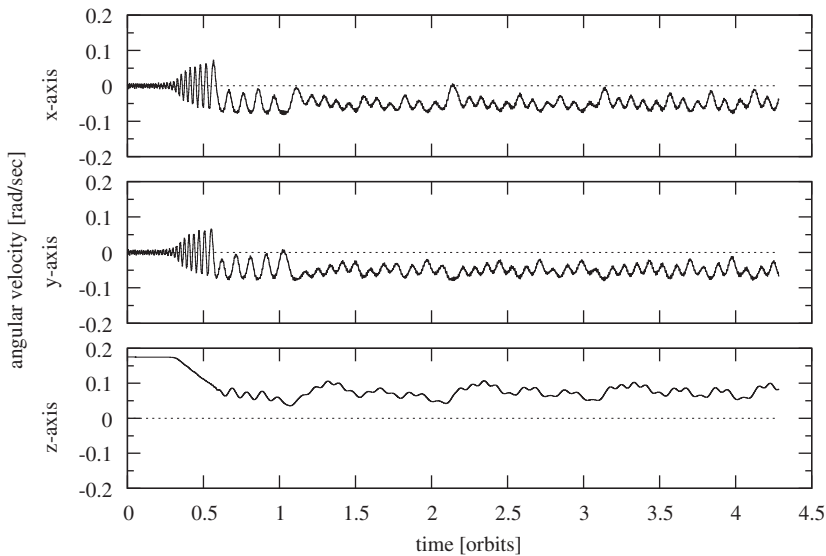


Fig. 20. Rotation z scenario.

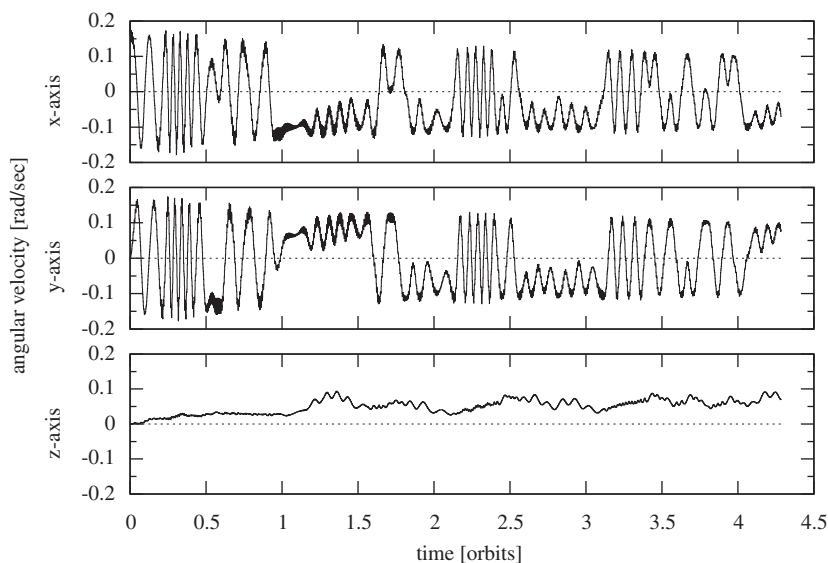


Fig. 21. Rotation x scenario.

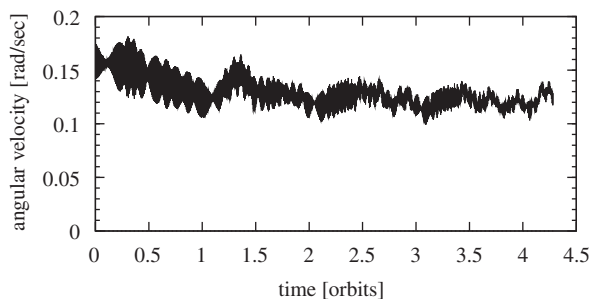


Fig. 22. Magnitude of the angular velocity vector, showing the small dampening effect.

to not align well with the magnetic field lines but mainly to dampen excessive rotational rates.

The magnetic alignment for the steady scenario is shown in Fig. 23. The axis of the permanent magnet is no longer perfectly aligned to the magnetic field lines. The deviation is about 80° . Small oscillations are present which are slowly damped.

The system response for rotating initial conditions is very similar (see Figs. 24 and 25). After one orbit, magnetic lock is achieved. Residual oscillations are damped.

The dampening effect is shown in Figs. 26 and 27.

6.1.3. Hysteresis correction

As it is stated in [15], the hysteresis material does not behave as theory predicts. A much weaker influence of the hysteresis material on the angular velocity is observed in experiments. For this reason a correction factor is introduced into the hysteresis model. The correction factor decreases the effective magnetic field strength of

the external field on the hysteresis material. This results in a lower magnetization of the material.

With a correction factor of 50 the area of the hysteresis cycle decreases significantly. The Earth magnetic field is no longer able to saturate the hysteresis material. Figs. 28–30 show the cycle of a hysteresis rod during simulation with a correction factor of 0, 50, and 100, respectively. The complete saturated loop in the first case is reduced to much smaller minor loops in the latter cases. This surely results in a weaker dampening effect of the hysteresis rods.

The corresponding depression in angular velocity can be seen in Fig. 31. These plots can be compared to the previous ones but have a much longer simulated duration of 26 orbits. The initial condition is the same as in the rotation-x scenario.

In the first plot a massive decay to about 0.02 rad/s can be seen over the first two orbits going into a very shallow decrease of $0.0136^\circ/\text{s}$ per orbit. The second graph shows an almost constant depression of $0.1089^\circ/\text{s}$ per orbit over the complete simulated time. The third graph shows a constant depression of $0.0145^\circ/\text{s}$ per orbit over the complete simulated time.

Some hypotheses on the different behavior of hysteresis material are:

- The magnitude of the hysteresis effect does not scale linear with the size and volume.
- The saturation of the material takes more time than the rotational change, creating more minor loops.
- The influence of other magnetic material in the satellite (e.g. the permanent magnet) on the hysteresis material is much higher.

These theories have to be evaluated in future work.

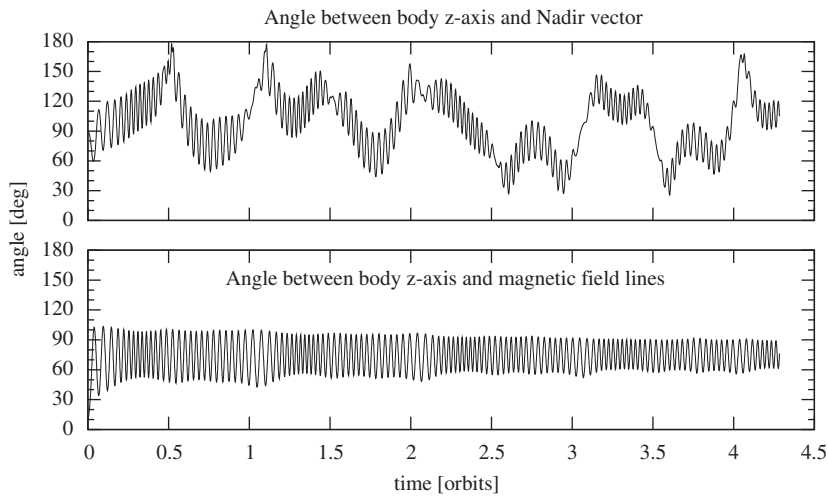


Fig. 23. Alignment for the steady scenario.

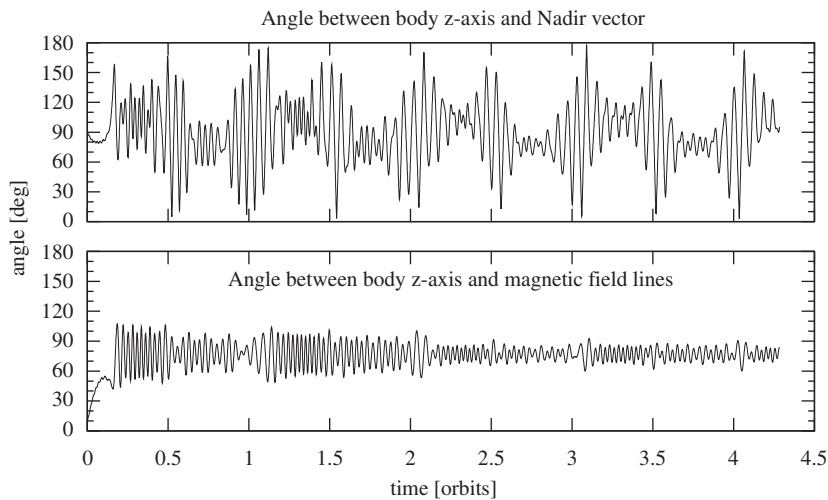


Fig. 24. Alignment for the rotation z scenario.

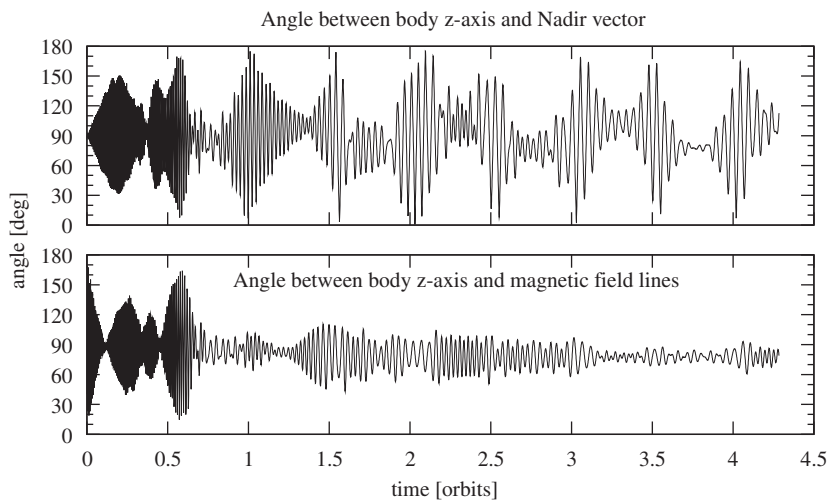


Fig. 25. Alignment for the rotation x scenario.

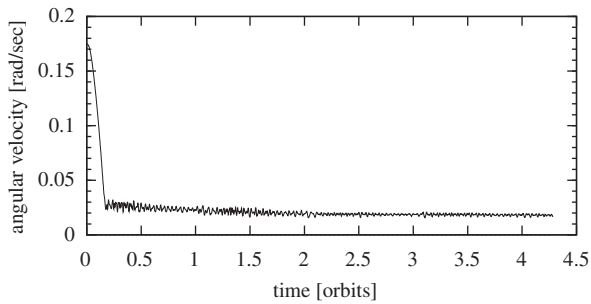


Fig. 26. Magnitude of the angular velocity vector for rotation z scenario.

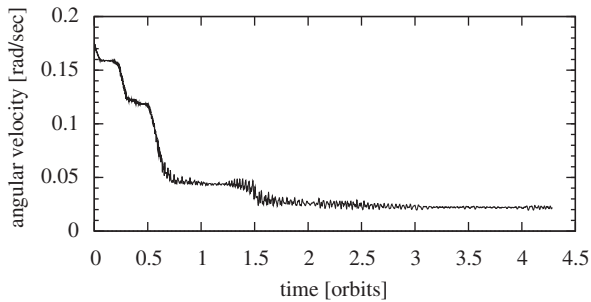


Fig. 27. Magnitude of the angular velocity vector for rotation x scenario.

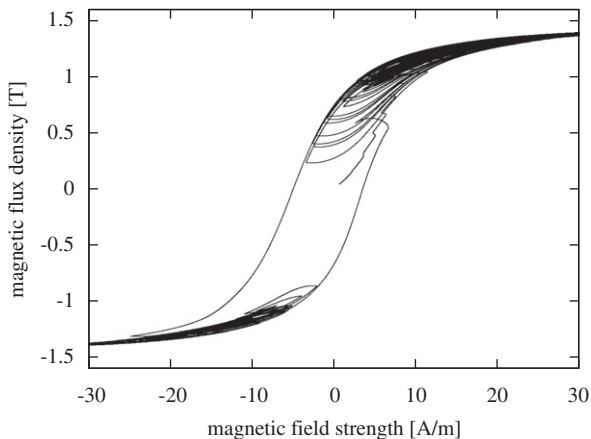


Fig. 28. Hysteresis curve with no correction factor. Saturation and minor loops can be seen.

6.2. Power behavior

The power generation is mostly independent on the magnetic stabilization configuration. Fig. 32 shows the generated power for a fast rotating scenario and a steady scenario. Both diagrams show an average power generation of about 3.2 W. This value is given for an efficiency of the voltage regulator of 90.8%. Taking into account a total efficiency of the power generation string of about 75%, the generated power can drop to 2.4 W. This might correspond to some power shortages registered on Delfi-C³ on orbit. Unfortunately the onboard power measurement of Delfi-C³ does not provide any information on the total availability or consumption of electrical power.

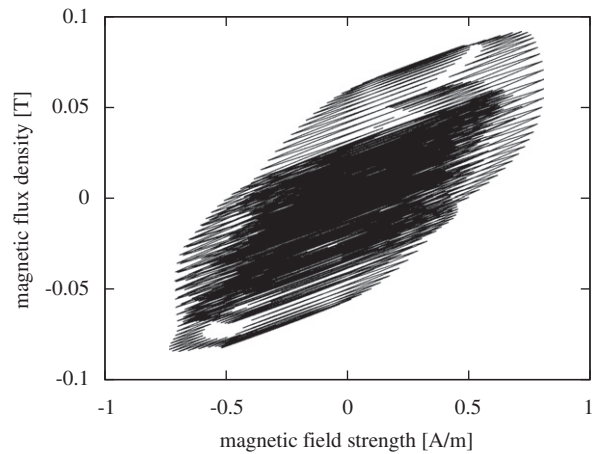


Fig. 29. Hysteresis curve with correction factor 50. Only minor loops are present.

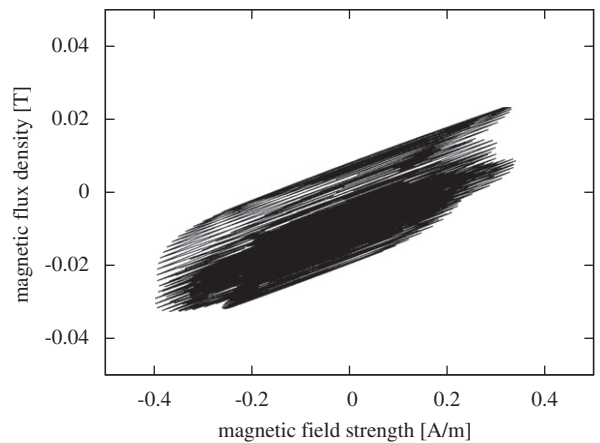


Fig. 30. Hysteresis curve with correction factor 100. Only minor loops are present.

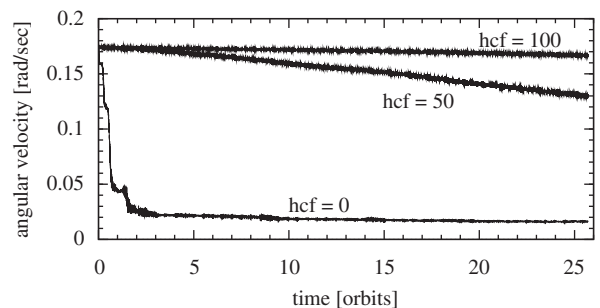


Fig. 31. Angular velocity depression with a hysteresis correction factor (hcf) of 0, 50 or 100.

7. Conclusion

System simulation can contribute to the development cycle of a satellite project. The presented modeling method utilizes a very modular and adaptive approach. There are many similarities to the concurrent engineering

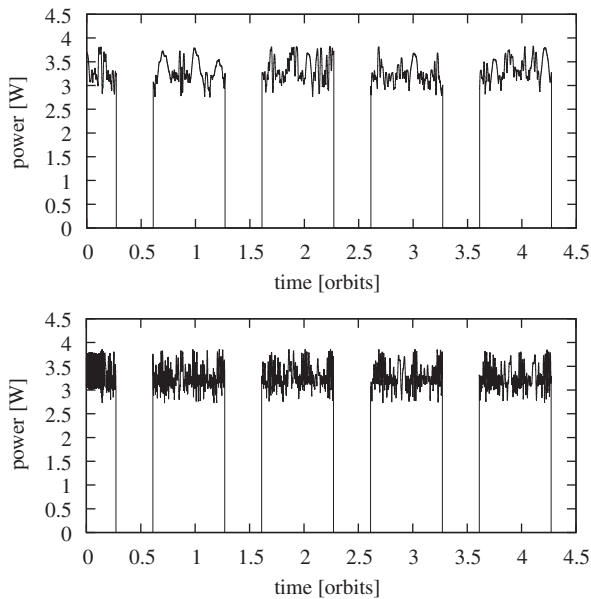


Fig. 32. Generated power for steady (top) and rotating (bottom) initial conditions.

approach. A central model of the system is present on which different engineering disciplines can work concurrently. The system model described here holds data on the structure and the dynamic behavior of the spacecraft system. With a growing library of building blocks one can easily create a system model out of the box and run dynamic simulations. The dynamic system simulation can be used to size critical design parameters, analyze the system response under varying conditions, and to verify the system performance.

In an application scenario, the complete approach to perform a system simulation was used. It has been shown that a simulation of a complete system is feasible and can be built by modular model building blocks. The dynamic attitude and power behavior of the nanosatellite project Defli-C³ has been simulated. The interactions between orbit, attitude and power behavior were included. The

next step is to complete the system model and include thermal and communication behavior.

However the attitude behavior of the satellite shows some discrepancies with real measurement. One has to keep in mind that a simulation is only as good as the model behind it. Some more effort has to be made to come up with a suitable hysteresis material model to improve the PMAS simulation of small satellites.

References

- [1] Object Management Group, OMG Systems Modeling Language (OMG SysMLTM). Version 1.1, 2008.
- [2] M. Raif, M. Brandstätter, C. Eckl, and U. Walter, A modular approach for modeling and dynamic simulation of spacecraft systems, in: Proceedings of the 7th IAA Symposium on Small Satellites for Earth Observation, 2009.
- [3] T. Coffee, The future of integrated concurrent engineering in spacecraft design, The Lean Aerospace Initiative Working Paper Series, Massachusetts Institute of Technology, October 2006.
- [4] O. Montenbruck and E. Gill, Satellite Orbits. Models, Methods, and Applications (Springer, Berlin, 2005).
- [5] J.R. Wertz, Spacecraft Attitude Determination and Control (Kluwer Academic Publishers, Dordrecht, The Netherlands, 1978).
- [6] D.A. Vallado, W.D. McClain, Fundamentals of Astrodynamics and Applications, vol. 12, Kluwer Academic Publishers, Dordrecht, 2001.
- [7] E. Gill, Smooth bi-polynomial interpolation of Jacchia 1971 atmospheric densities for efficient satellite drag computation, DLR-GSOC IB (1996) 91–96.
- [8] J. Davis, Mathematical modeling of Earth's magnetic field, Technical Note, 2004.
- [9] W.J. Larson and J.R. Wertz, Space Mission Analysis and Design. Third edition (Microcosm Press, El Segundo, Calif., 1999).
- [10] J.-F. Levesque, Passive magnetic attitude stabilization using hysteresis materials. SIGMA-PU-006-UdeS, 2003.
- [11] J. Tellinen, A simple scalar model for magnetic hysteresis, IEEE Transactions on Magnetics 34 (1998) 4.
- [12] M.M. Schiffner, M. Brandstätter, U. Walter, A generalized system architecture tool for concurrent design, 2nd Concurrent Engineering for Space Applications Workshop 2006, ESTEC, Noordwijk, The Netherlands, 2006.
- [13] W. Ubbels, et al., First flight results of the Delfi-C³ satellite mission, 22nd Annual AIAA/USU Conference on Small Satellites (2008).
- [14] F. te Hennepe, B.T. Zandbergen, R. Hamann, Simulation of the attitude behaviour and available power profile of the DELFI-C³ spacecraft with application of the OPSIM platform, in: 1st CEAS European Air and Space Conference, Berlin, Germany, 10–13 September 2007.
- [15] F. Poppenk, in: Design and testing of an attitude control system for a nano satellite, Master Thesis Report, Delft University of Technology, 2009.